

Apache Spark

Alka Bhushan

Apache Spark

- A distributed computing platform
- Can handle wide range of workloads such as batch applications, iterative algorithms, interactive queries and streaming

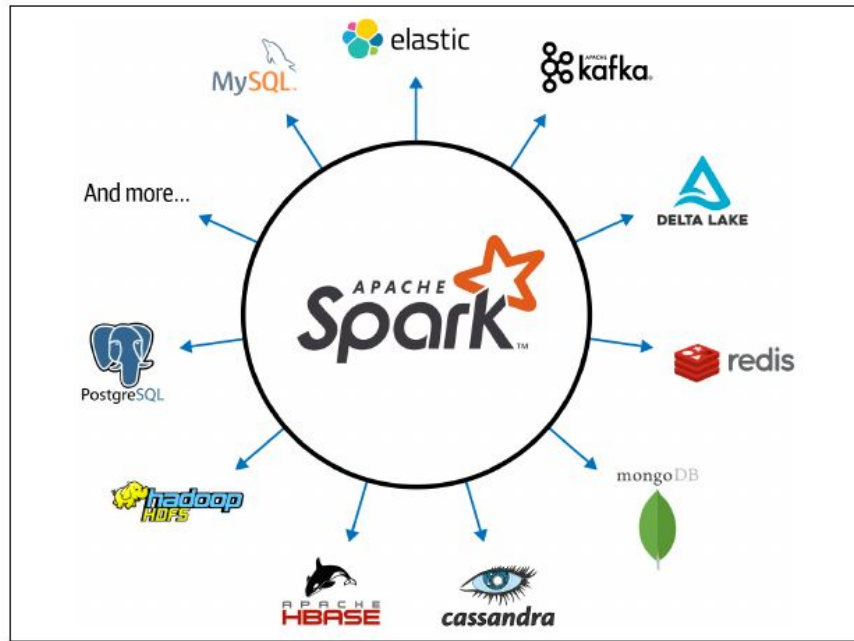
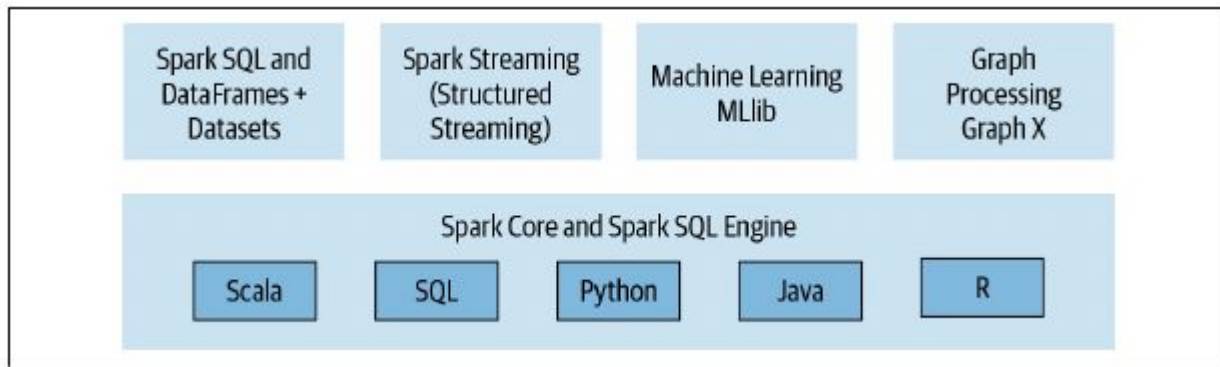


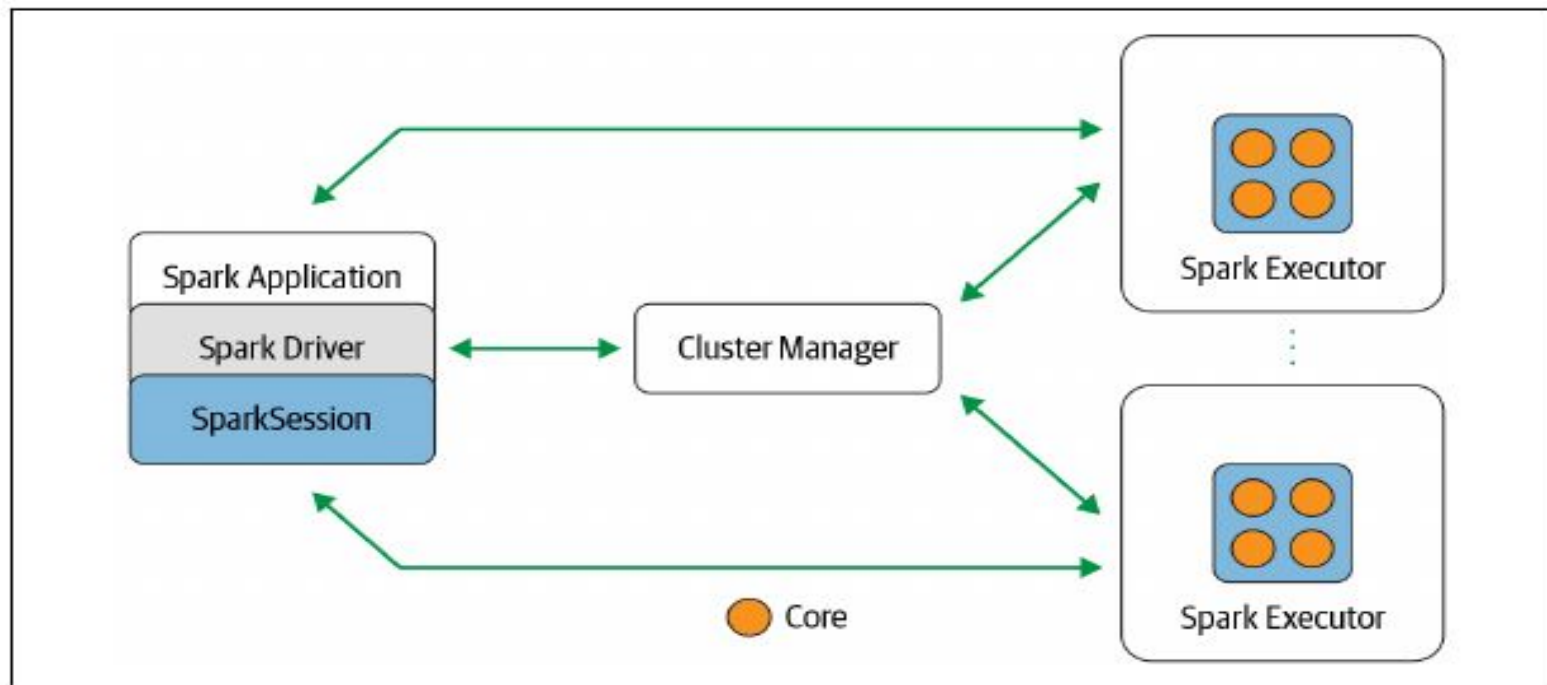
Figure: Spark's ecosystem of connectors

Spark Components

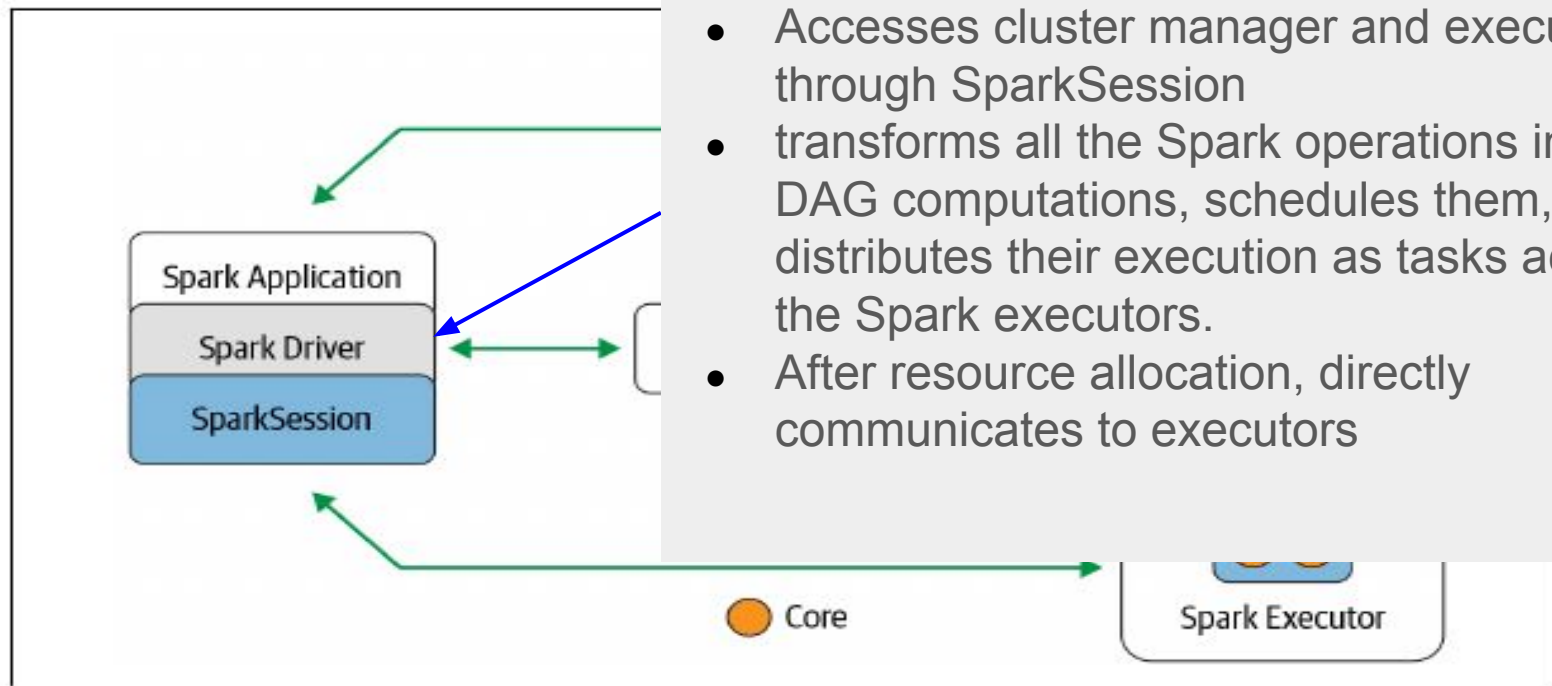


- **Spark SQL** : RDBMS Table, CSV, PARQUET, JSON etc.
- **MLlib** : ml algorithms, linear algebra operations and statistics, low level ml primitives (gradient decent optimization), feature extraction, transformation, pipelines, model tuning
- **Streaming** : real time streaming
- **Garph X**: graph algorithms - page rank, connected components, triangle counting etc.

Spark Distributed Execution



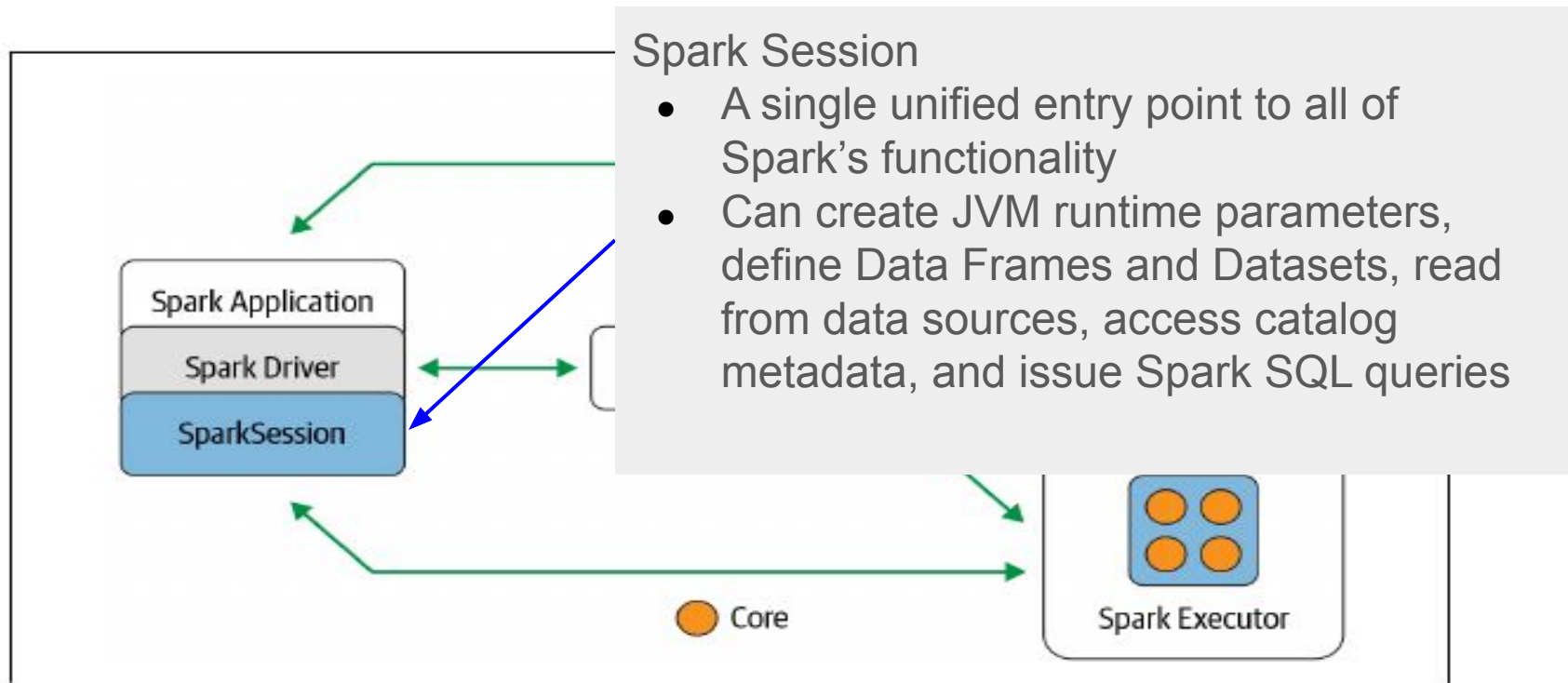
Spark Distributed Execution



Spark Driver

- Creates spark session object
- Accesses cluster manager and executors through SparkSession
- transforms all the Spark operations into DAG computations, schedules them, and distributes their execution as tasks across the Spark executors.
- After resource allocation, directly communicates to executors

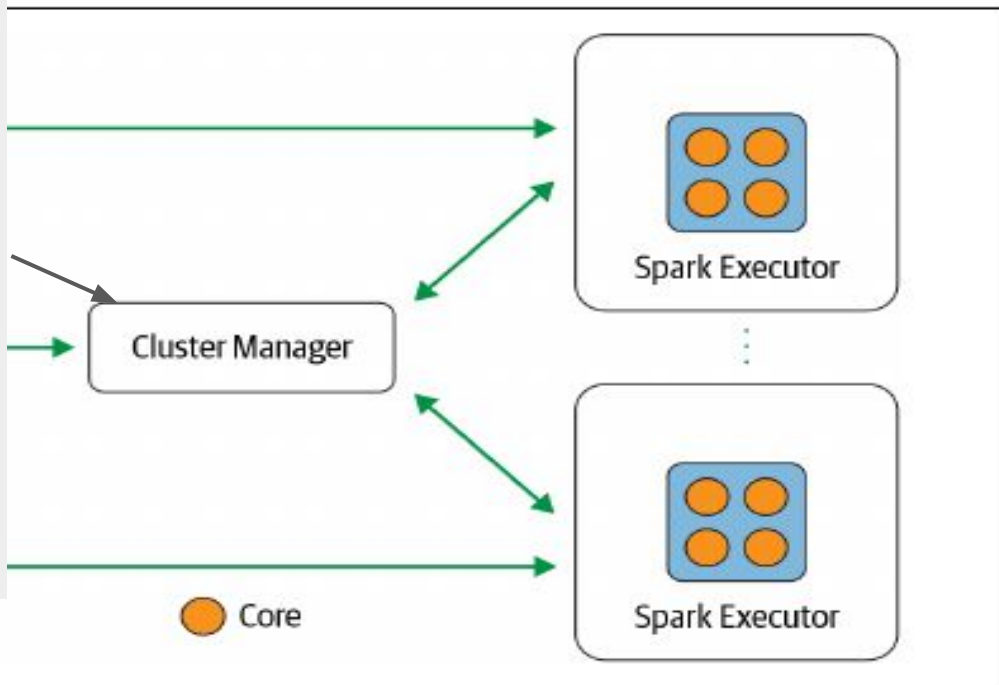
Spark Distributed Execution



Spark Distributed Execution

Cluster Manager

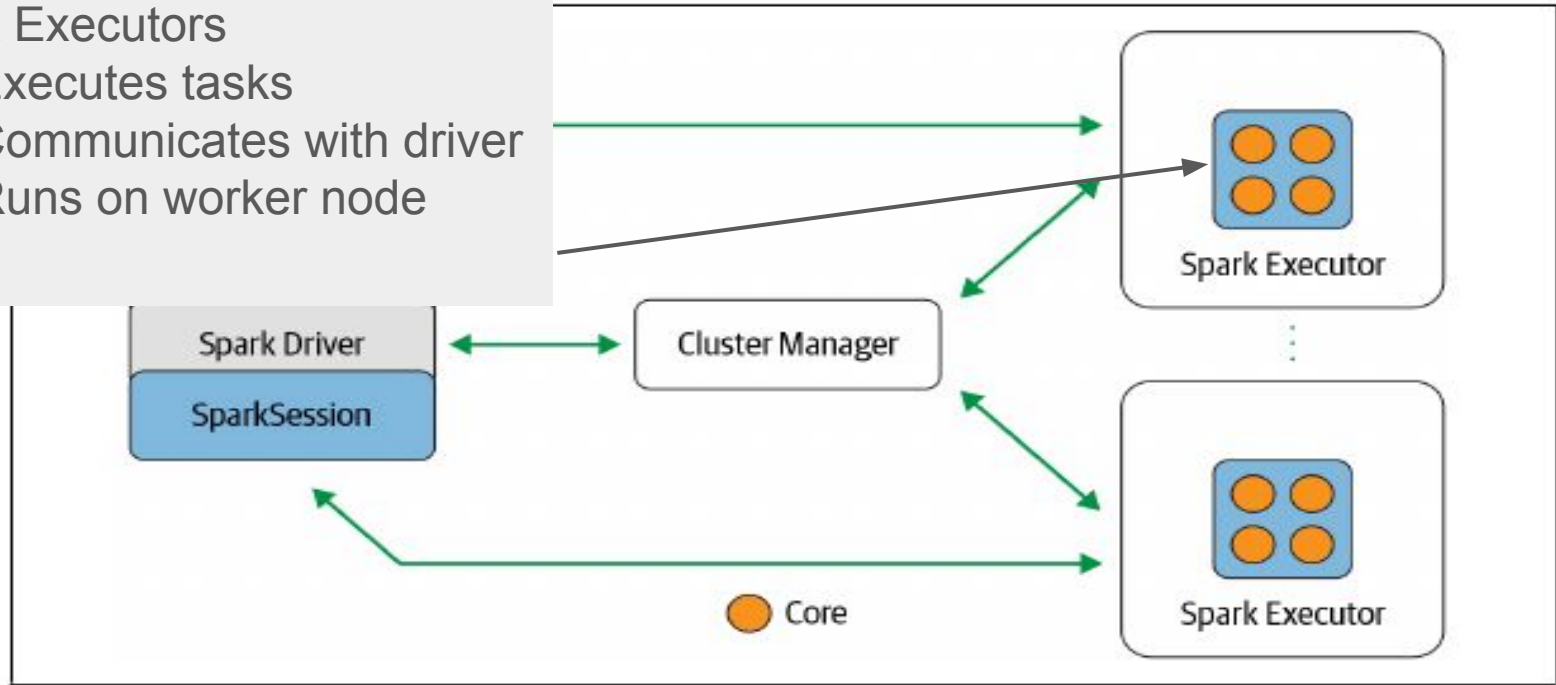
- responsible for managing and allocating resources for the cluster of nodes on which your Spark application runs.
- Local, standalone, Yarn, Kubernetes



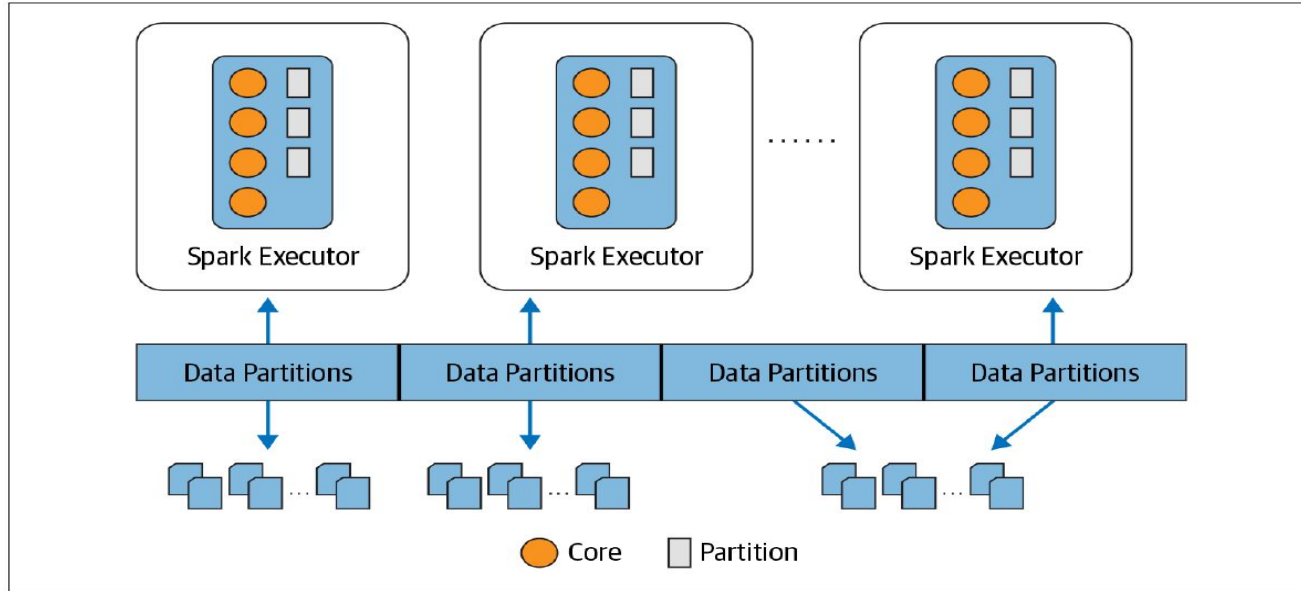
Spark Distributed Execution

Spark Executors

- Executes tasks
- Communicates with driver
- Runs on worker node



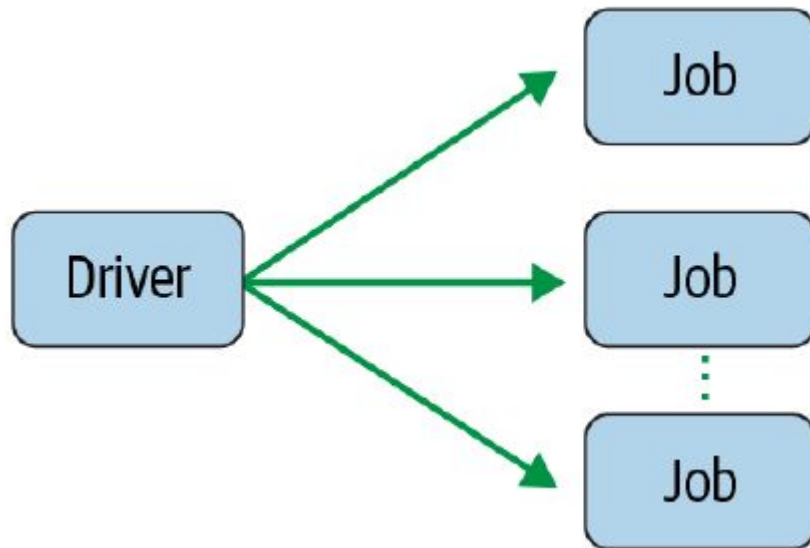
Distributed Data and Partitions



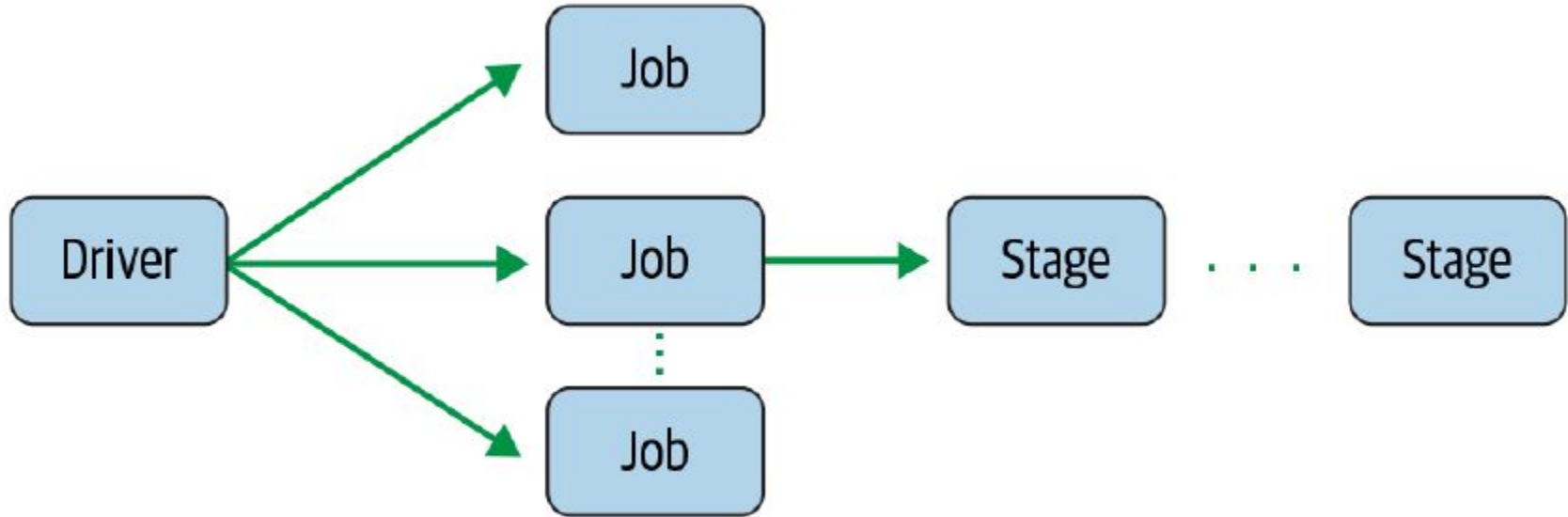
- Each executor gets more than one partition
- Number of partitions should be more than number of executor cores

Job

- More than one job in a single application
- A DAG is created for each job
- Triggers by action (e.g. `collect()`, `count()`, `show()`)
- `collect()` brings all data into memory
 - Should be run with caution



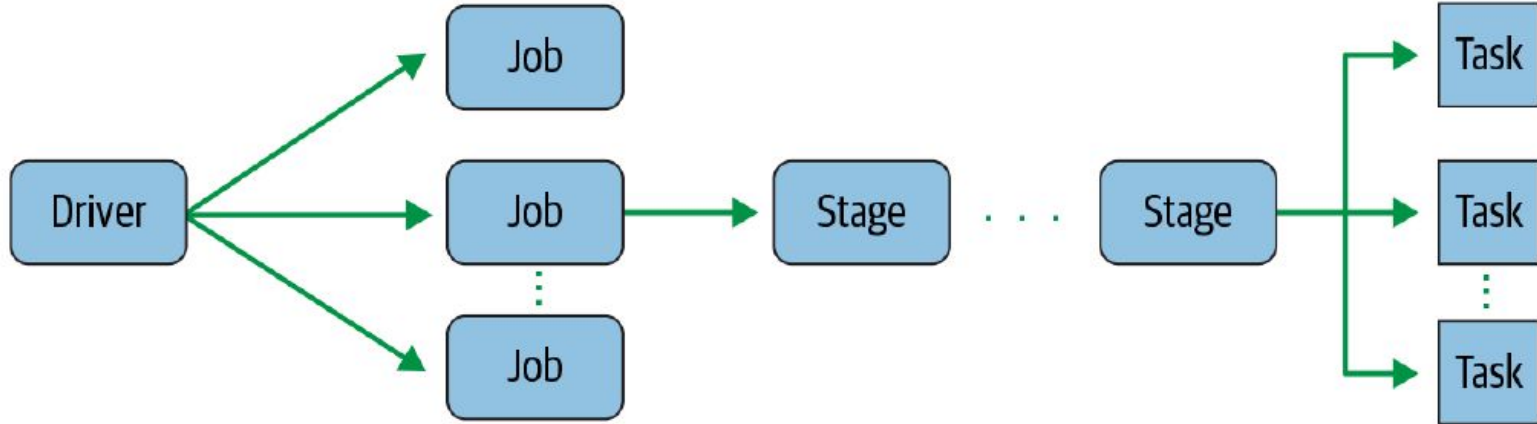
Stage



- Stages are created for each shuffle operation
- `groupByKey()`, `reduceByKey()`
- Shuffle operations are usually more expensive than action operations

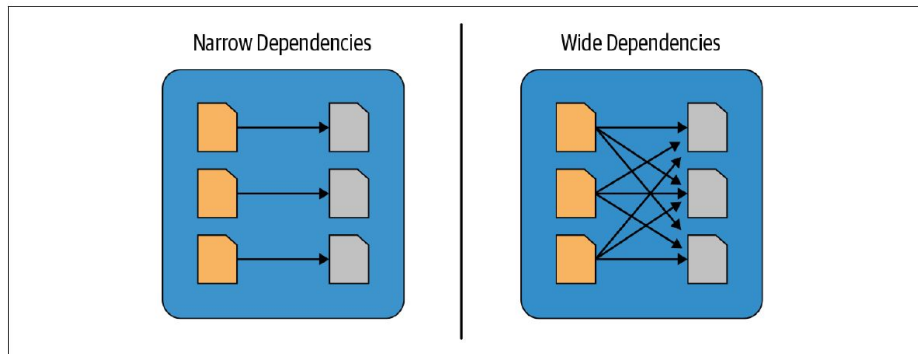
Task

- All tasks in a stage run in parallel
- A task is assigned to a partition
- A task is a set of steps



Spark Operations

- Action : cause data to be materialized
 - Return to driver , or
 - Save to file
- Transformations (lazy) : Narrow and Wide
- Wide transformation causes shuffling of data between executors



Transformations	Actions
<code>orderBy()</code>	<code>show()</code>
<code>groupBy()</code>	<code>take()</code>
<code>filter()</code>	<code>count()</code>
<code>select()</code>	<code>collect()</code>
<code>join()</code>	<code>save()</code>

Data Collections Strategies and Structured APIs

Structured APIs

- Spark dataframe
- Spark SQL

Data Collection Strategies

- Batch (files, database, cloud space),
- Streaming (kafka),
- Batch +Streaming (hybrid)